## Week 5 | Data Visualisation: Design Choices Are Arguments

Reflection This week, our group continued working with the dataset we gathered for our project on AI-assisted reading. We collected discussion data from online forums, and many of the platforms we used came with built-in automatic visualisations. At first glance, these ready-made charts felt objective and straightforward, as if they simply reflected what people thought about AI reading tools. Yet as we cleaned and filtered our raw data, I began to realise that every rule we set for data cleaning would shape the final visual output. The thresholds we chose for including or excluding posts, how we tagged different kinds of comments, and the way we grouped discussion themes were not neutral technical decisions. These choices, long before we produced any final chart, were already narrowing the story the visualisation could tell.

This aligns closely with Feigenbaum and Alamalhodaei's (2020) discussion of visual data storytelling. They argue that data visualisation is never just a window onto pre-existing facts; it is a form of argument that shapes how audiences understand the subject. In our case, the automatic visualisations provided by the forum platforms carry this same ideological work. They draw attention to certain patterns while hiding others, and they make these constructed patterns appear natural and self-evident. Even before we made our own visual design decisions, the way the data had been captured, filtered and categorised by the platform already guided what could become visible.

Our dataset consists of voluntary public discussions about AI-supported reading. This raises a further critical problem. Forum posts only capture a small, self-selected subset of people's views. We have no reliable way to verify whether these comments represent wider attitudes, or even whether individual posts are sincere. The visualisations, whether auto-generated by the site or created by our group, can only serve as partial reference material. It would be methodologically unsound to treat them as proof of broader public opinion or general conclusions about AI reading practices. A clean, polished visual has the power to make limited, uncertain data look authoritative, which is a risk I had not fully considered before this workshop.

Thinking with D'Ignazio and Klein's (2020) feminist data principles, I also considered what a more responsible visualisation might look like here. Instead of smoothing out uncertainty to produce a neat chart, a feminist approach would make the limits of the data explicit. It would mark where the sample is narrow, where verification is impossible, and where the boundaries of our evidence lie. Too often, visualisation design prioritises clarity and coherence at the cost of hiding uncertainty and sampling bias. The attractive, easy-to-read chart can quietly erase its own limitations.

I also read one of the data stories published in The Pudding. It helped me see how skilled visual storytelling can communicate nuance, but it also reminded me that even thoughtful visuals still rely on prior decisions around sampling and filtering. A strong

visual narrative does not pretend its data is complete or universally representative; it makes its constraints part of the story.

This week taught me that data visualisation is embedded within the whole research pipeline. The choices begin at the moment we define what counts as usable data, not only when we pick colours or axes for a chart. For digital media researchers, it is not enough to interpret what a visual appears to show. We must trace backwards: what was excluded during cleaning? Who is and is not represented in this sample? What uncertainties cannot be resolved?

I have added screenshots of the platform's native visualisations alongside notes on our data-cleaning criteria. I also wrote down the limits of our forum dataset: its narrow sample and the difficulty of verifying the authenticity of user contributions.

